**Practical No.: 8**

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**Problem Statement:** Maximise Profit by Shipping Partial Orders (Fractional Knapsack). You run a shipping company and need to load a truck with parcels of different weights and profits. The truck has a limited weight capacity. Write a program to choose parcels (even partially) to maximise profit using the Fractional Knapsack strategy.

**Code:**

```
# Fractional Knapsack using Greedy Algorithm
n = int(input("Enter number of parcels: "))
items = []
for i in range(n):
  print(f"\nParcel {i + 1}")
  weight = float(input("Enter weight: "))
  profit = float(input("Enter profit: "))
   if weight <= 0:
    print("Weight must be greater than zero.")
    exit()
   ratio = profit / weight
  items.append([weight, profit, ratio])
capacity = float(input("\nEnter truck capacity: "))
if capacity < 0:
  print("Capacity cannot be negative.")
  exit()
```

```
# Sort by profit/weight ratio in descending order
items.sort(key=lambda x: x[2], reverse=True)
totalProfit = 0
print("\nSelected Parcels:")
for item in items:
   weight = item[0]
   profit = item[1]
 if capacity == 0:
       break
if weight <= capacity:
       print("Full Parcel -> Weight:", weight,
            "Profit:", profit)
       capacity -= weight
       totalProfit += profit
else:
       fraction = capacity / weight
      print("Partial Parcel -> Weight:", round(capacity, 2),
           "Profit:", round(profit * fraction, 2))
      totalProfit += profit * fraction
       capacity = 0
print("\nMaximum Profit =", round(totalProfit, 2))
```

**Output:**

Enter number of parcels: 4

Parcel 1

Enter weight: 20

Enter profit: 45

Parcel 2

Enter weight: 45

Enter profit: 30

Parcel 3

Enter weight: 67

Enter profit: 10

Parcel 4

Enter weight: 89

Enter profit: 34

Enter truck capacity: 100

Selected Parcels:

Full Parcel -> Weight: 20.0 Profit: 45.0

Full Parcel -> Weight: 45.0 Profit: 30.0

Partial Parcel -> Weight: 35.0 Profit: 13.37

Maximum Pro**fit = 88.37**